

08 — PTI 3250: the Discharge Gate Passes Silently

Proves · The PTI discharge gate passing silently: enough shaft-motor capacity, no change to any result.

Mechanics this scenario turns on

- PTI is the shaft motor: the battery pushing power through the main engine's shaft. A gate checks that the installed PTI capacity can carry the propulsion band the battery is asked to shave; if not, the whole calculation is refused with a 400.
- Cascade per row: H = what it wants · $I = \min(\text{remaining budget}, H)$ · $J = I \times \text{CoverageFactor}$ (covered) · $L = H - J$ (left to the gensets). Priority: DpReserve → DpDemand → Mission → Propulsion → Hotel. Invariant $\Sigma J + \Sigma L = \Sigma H$ — the sea sets the swing, the battery moves the split.

Panels described below · What the PTI field switches on · The gate arithmetic · Result panels

Anything not described here behaves exactly as in `01-excel-baseline` — same plant, same hours; only what this scenario changes is worked through below.

Trust · verified against the reference workbook. These figures are proof.

Read after · scenario 01.

Test 01 + **PTI Capacity per Main Engine = 3 250**. Everything else identical.

What the PTI field switches on

With PTI = 0/empty the battery is modelled "at switchboard level" (no feasibility question, ADR-5). A positive PTI enables the Excel-fidelity check: the battery's covered PROPULSION band must physically reach the shaft through the shaft machines (motor mode).

The gate arithmetic (per combination)

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PTI capacity    = installed MEs × MaxPti = 2 × 3 250 = 6 500 kW    (union rule, decision D5)
PTI used for propulsion assist = 0      (ME 2×24 000 has no deficit at 11 836 kW)
Headroom        = 6 500 - 0 = 6 500
Required        = battery's propulsion covered band = 200.6 kW

6 500 ≥ 200.6 → every combination passes → nothing is filtered
```

Result panels

Identical to test 01 in every number (SR 444.7 / PS 204.4, same 5 combos, baseline 13 307.4, IL1 13 286.2, benefit 173.7). The gate is a guard, not a feature — a wide-open door changes nothing. That silence IS the test result.

Takeaway: paired with test 09 (same field = 50 → hard 400 error), this pins both sides of the gate's threshold. Manual boundary if curious: 100/engine still fails, 101/engine passes ($2 \times 101 = 202 \geq 200.6$).

